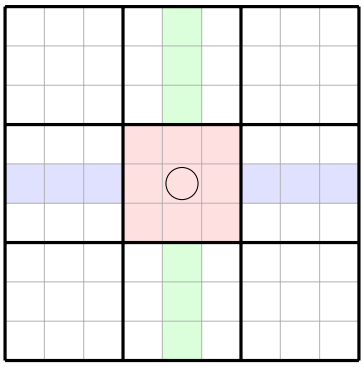
# Sudoku

## 1 The puzzle and its rules

A sudoku grid is a $9 \times 9$ board of cells, subdivided into nine $3 \times 3$ *boxes*. A puzzle starts with some cells filled in (the *givens*); the task is to fill every remaining cell with a digit from 1 to 9 such that

- every **row** contains each digit exactly once,
- every **column** contains each digit exactly once,
- every **box** contains each digit exactly once.

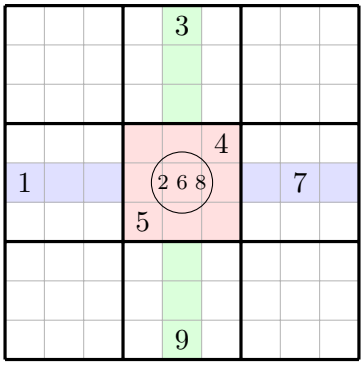
Each cell therefore belongs to three *units* (its row, its column, its box), and its 20 distinct *peers* — the other cells of those units — can never share its digit.



The three units of the circled cell: its row, its column, its box.

## 2 Candidates and constraint propagation

The solver's central data structure attaches to every empty cell its *candidate set*: the digits not yet ruled out. A digit placed in a cell is erased from the candidates of all 20 peers. In the position below, the circled cell sees $1, 7$ in its row, $3, 9$ in its column and $4, 5$ in its box, leaving candidates $\{2, 6, 8\}$ — the small digits a pencil-and-paper solver would write in the corner:




Elimination: $\{1, \ldots, 9\}$ minus the row, column, and box gives $\{2, 6, 8\}$.

Eliminations feed two deduction rules, and the rules feed further eliminations:

- **Naked single.** A cell whose candidate set shrinks to one digit must hold that digit. Placing it erases the digit from 20 more candidate sets — which may create new singles.
- **Hidden single.** If, within one unit, some digit remains possible in only one cell, that cell must hold it — even if the cell still has other candidates. Below, the digits $2, 4, 7$ are missing from the row; 4 appears in only one candidate set, so it goes there:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 **4** 7 | 3 | 5 | 2 7 | 6 | 2 7 | 8 | 9 |

A hidden single: only the second cell can still take the 4.

Running eliminations and both rules until nothing changes is *constraint propagation* — computing a fixpoint. Its power is easy to underestimate: most published easy and medium puzzles dissolve completely under propagation alone, with never a guess.

## 3 Search: when logic stalls

On hard puzzles propagation reaches a stable position with unresolved cells. The solver then guesses: pick an unresolved cell, try one of its candidates, propagate, and recurse; a contradiction (some cell left with no candidates) refutes the guess, and backtracking tries the next one. This is the familiar search tree — with two twists that keep it tiny compared to the tours and queens of earlier quizzes:

- **Choose the most constrained cell** (fewest candidates remaining), the *minimum remaining values* rule — the same rescue-the-scarce principle as Warnsdorff's onward counts. A cell with two candidates gives a wrong guess a 50% chance of instant refutation.
- **Propagate after every guess.** Each guess triggers the full cascade, so consequences surface immediately and contradictions appear after a handful of placements, not deep in the tree.

## 4 Facts worth knowing

- **Uniqueness.** A *proper* puzzle has exactly one solution; verifying properness means searching for a second solution and failing. A puzzle generator must run exactly that check on every candidate puzzle it digs out.
- **Seventeen givens.** No proper 9x9 puzzle with fewer than 17 givens exists (McGuire, Tugemann and Civario, 2012 — by exhaustive computer search), and 17-given puzzles do exist.
- **It scales badly, provably.** On generalized $n^2 \times n^2$ boards, sudoku is NP-complete; the 9x9 case is only tamed by its fixed size and by how sharply propagation prunes.

## 5 From mathematics to program

The program keeps the grid as 81 candidate sets (a solved cell is a singleton) and implements propagation as the fixpoint of the two rules; the solver wraps it in MRV-guided backtracking. The renderer draws what solvers on paper actually do: pencil marks shrinking as eliminations cascade, deduced digits appearing in one color, guesses in another, and a contradiction flashing before its branch rewinds — constraint propagation made watchable.